

Timothy S. Gilbert

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[Portfolio](#) · [GitHub](#) · [Desert Data Labs](#)

PROFESSIONAL SUMMARY

Ecologist, data engineer and founder of Desert Data Labs LLC, with 8+ years across field ecology, research data management, and full-stack data application development. Specializes in R Shiny applications, SQL database management, and ETL pipelines. Completing an M.S. in Data Science at the University of Arizona.

RELEVANT EXPERIENCE

Founder & Principal Data Engineer

Desert Data Labs LLC: Tucson, AZ — 2026–Present

- Founded an Arizona LLC delivering custom dashboards, databases, and data pipelines for small businesses, nonprofits, and research teams without in-house software support
- Design and build production R Shiny applications, SQL databases (Azure SQL, PostgreSQL, SQL Server, SQLite), and ETL/integration pipelines with embedded QA/QC
- Developing EcoPlot Mobile, an offline-capable field-data collection web app (PWA) on Azure (Static Web Apps + managed SQL), as a modern successor to legacy desktop field tools ([click for demo](#))
- Build and maintain the company marketing site in Quarto with automated CI/CD deployment via GitHub Actions
- Established the service catalog, fixed-scope SOW process, hosting/maintenance tiers, and a sliding-scale “Mission Pricing” program providing reduced-rate and pro bono data infrastructure to under-resourced ecology and conservation labs

IT Support Computing Analyst I–III

University of Arizona, VGS Project: Tucson, AZ — March 2022–Present;

promoted from Analyst I to Analyst III Nov 2022, full time

- Developed and deployed R Shiny applications for research data workflows, including data ingestion, management, manipulation, quality assurance, and interactive visualizations
- Designed ETL pipelines for historical ecological datasets, parsing Excel files into structured SQL tables with QA/QC protocols built in
- Maintained local and server-side databases (Microsoft SQL Server, SQLite, Azure Data Studio, MySQL); automated SQL scripts for data quality and reporting
- Conducted 20+ training sessions for USFS and NRCS staff at field offices, demonstrating VGS platform workflows, instructing on data collection best practices, and delivering technical support and field-device maintenance
- Led and mentored a student data entry team, standardizing QA protocols for high-volume historical data transcription
- Supported cross-platform software testing (Windows, iOS, Android) in a C# environment using Visual Studio and VS Code; tracked bugs and fixes via Jira
- Supported agile development workflows and sprint tasks across technical and field teams; led daily scrum meetings
- Assisted in preparing quarterly reports for federal funding sources, documenting project progress, deliverables, and achievements

Rangeland Specialist

University of Arizona, Gila County Cooperative Extension (Rangeland Monitoring): Tucson, AZ — Oct 2017–Apr 2022, seasonal (Sept–Apr), full time

- Conducted long-term vegetation monitoring across rangeland sites using VGS protocols (line intercept, plant diversity, cover, biomass production)
- Provided testing feedback and field validation for VGS platform development, improving usability and data accuracy, eventually leading to a role on the VGS team (above)
- Supported ecological data collection and pasture assessments within the Tonto National Forest in coordination with federal land managers
- Trained USFS employees and local ranchers in standardized plant data collection methods; facilitated collaborative monitoring efforts
- Managed data entry for historical data
- Led seasonal field staff in vegetation monitoring efforts

Lead Aquatics Field Ecologist

National Ecological Observatory Network (NEON): Tucson, AZ — March 2018–Nov 2021, seasonal (Mar–Dec), full time

- Led field sampling campaigns across aquatic and terrestrial systems (water chemistry, invertebrates, flora, and fauna) at Sycamore Creek, Santa Rita Experimental Range, and Jornada Experimental Range
- Collected water-quality metrics, microbial DNA, stream discharge measurements, electrofishing data, and sediment profiles; processed fish vouchers and soil samples for downstream identification and analysis
- Executed terrestrial protocols including small mammal trapping (tagging, biometrics, hair, blood, and ear samples), ground beetle surveys, vegetation structure assessments, and litter trap deployment
- Performed laboratory analysis under sterile conditions, titrations, and reagent preparation; cleaned, sterilized, and maintained lab glassware and equipment
- Contributed to drafting and revising NEON Standard Operating Procedures (SOPs) and followed strict protocol guidelines to ensure data quality
- Managed inventory of field and lab equipment, maintained accurate supply records, and placed orders to prevent stockouts
- Developed R scripts for QA/QC of ecological datasets after field work, improving data integrity and workflow efficiency

Fuels Monitoring Technician

Great Basin Institute (GBI) / Bureau of Land Management: Ely, NV — Jun 2017–Oct 2017, full time

- Conducted vegetation and fuels assessments using Assessment, Inventory, and Monitoring (AIM) protocols
- Delivered field data for fire ecology and fuels management programs across BLM-managed lands

EDUCATION

University of Arizona - M.S. Data Science (Online), Expected Dec 2026

University of Arizona - B.S. Natural Resources: Wildlife Conservation & Management, May 2017

PROGRAMMING PROJECTS

Where Do Big 12 Recruits Come From?

An R Shiny app to visualize Big 12 football and basketball recruiting classes — exploring where recruits came from (distance from high school to college destination) on a map, and plotting distance traveled by recruits over time. Select a team to get started. [View project ↗](#)

Small Mammal Tracker

An R Shiny app using National Ecological Observatory Network data to compare small mammal capture data across NEON sites by location and date range, ranking sites by total captures and tracking individuals by tag to examine patterns and capture history, with site-level summary plots. [View project ↗](#)

Which State's Flora Matches Each Movie World Best?

An R Shiny app that builds chord diagrams showing shared species between selected U.S. states and movie worlds — Dune (Arrakis), Lord of the Rings (Middle-Earth), and Jurassic Park (Isla Nublar) — plus gauge plots apportioning a state's flora across movie-themed biomes. [View project ↗](#)

VGSLite

An R-based helper application (.exe installer) built to support VGS 5 Desktop on Windows devices, built in R and packaged with Electron. [View project ↗](#)

NCAA Sample Data Visualization

Scripts to scrape recruiting data from multiple sports sites (e.g., 247Sports, On3) and visualize Big 12 football and basketball class rankings over time. [View on GitHub ↗](#)

VGS Batch Importer

An ETL pipeline built in R to ingest historical Excel datasheets into a SQLite database, supporting multiple vegetation protocols (point ground cover, line intercept, nested frequency) across transects and sites. Parses data by key identifiers (e.g., SiteID), organizes metadata, and inserts structured records into a local VGS database, with embedded QA/QC checks to prevent corrupt or incomplete ingestion. [View on GitHub ↗](#)

REFERENCES

Donald E. Merson

Data Scientist Manager III
Web Applications Developer IV — VGS project (University of Arizona)
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